Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/03/2016
PI Name:	Wyatt, Sarah E Ph.D.		
Project Title:	Proteomics Analysis of Arabidopsis Seedlings in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	45701	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS Rapid Turn Around Flight NOTE: End date is now 8/31/2017 per NSSC information and PI (Ed., 10/3/16) NOTE: End date is now 8/31/2016 per NSSC information and PI (Ed., 10/8/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Luesse, Darron Ph.D. (Southern Illinois University, Edwardsville)		
Grant/Contract No.:	NNX13AM48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space flight environment provides a unique environment to understand how gravity informs plant growth, development, and physiological processes. The central objective of this study is to determine what proteins are differentially expressed during space flight in developing Arabidopsis seedlings. A body of research has evaluated the differential gene expression on space flight vs ground controls, but gene expression provides an indirect measure of what proteins are expressed. Genes may be transcribed but the transcripts degraded, or translated proteins may be regulated through post translational modifications. Our hypothesis is simple: Some components of plant physiology that are informed by gravity are regulated by post transcriptional or post translational mechanisms. Proteomic analysis is the best/only method to identify these components. We propose to use BRIC-PDFU (Biological Research In Canisters - Petri Dish Fixation Unit) hardware on a Rapid Turn-Around Space Flight Experiment to gain insights into differences in protein profiles between Arabidopsis seedlings grown during space flight and ground based controls. Arabidopsis seedlings will be germinated, grown, and tissues fixed on orbit. Once returned to Earth, total protein will be extracted, labeled with iTRAQ reagents, and analyzed for differences in protein profiles. These data will complement the gene expression data currently available from space flight experiments and also provide novel insights by supplying data on differences in post-transcriptional regulation. They will also provide a foundation for a network analysis to generate additional hypotheses as to the mechanisms involved in gravity perception/response in plants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research may lead to the discovery of novel components that determine how plants respond to gravity and to the space flight environment. It will also form a foundation for the establishment of hypotheses to support research on plant improvement for space flight and life support in the space flight environment.
Task Progress:	At the end of last year, we had extracted the RNA and proteins and had sent them to facilities for processing. In the past year, we have made substantial progress. Just to reiterate, roughly 20,000 Arabidopsis Wild Type Col-0 seeds were flown aboard the International Space Station using PDFU-BRIC hardware. After approximately 3 days, seedlings were fixed using RNAlater® and frozen at -80oC until return to Earth. A duplicate set of WT Col-0 seeds was maintained at Kennedy Space Center, also in PDFU-BRIC hardware, as a ground control. Two additional sets of seedlings, not in PDFU-BRIC hardware but in enclosed containers, were maintained at Ohio University under similar environment conditions as hardware and fixative controls for the space flight samples. Proteins were analyzed using iTRAQ 8-plex tandem mass spectrometry at Danforth Plant Science Center, St. Louis, MO. RNAseq was performed at Washington University, St. Louis, MO. RNA and protein data was returned in the fall of 2015, and we began analysis. In space flight and ground controls, 2641 soluble and 2917 membrane proteins were identified using Mascot software (Matrix Biosciences) and the TAIR10 database. The fixative, RNALater, had no detrimental effect on protein quality or identification. Analysis of hardware controls revealed 1651 differentially expressed transcripts, 108 differentially abundant soluble, and 39 membrane proteins. Analysis of samples preserved with and without RNAlater revealed 2750 genes differentially expressed and 84 soluble and 120 membrane proteins. Fold enrichment analysis of fixative controls show evidence of desiccation stress on the transcriptome. Filtering the space flight data for those transcripts and proteins expressed in fixative and hardware controls revealed 571 transcripts, 59 soluble proteins, and 36 membrane proteins deferentially expressed only as a result of space flight. Further analysis will help reveal the processes/mechanisms affected by space flight.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Basu P, Luesse DR, Wyatt SE. "A Discovery-based, Proteomic Analysis of Arabidopsis Seedlings Grown in Microgravity." Presented at Plant Biology 2015, Minneapolis, MN, July 26-30, 2015. Program: Plant Biology 2015, Minneapolis, Minnesota, July 26-30, 2015. , Jul-2015
Abstracts for Journals and Proceedings	Wyatt SE, Luesse DR. "BRIC-20: Proteomics analysis of Arabidopsis seedlings in microgravity." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Program:31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Kruse CPS, Luesse DR, Wyatt SE. "Transcriptomic Analysis of Seedlings Grown in Microgravity." Presented at 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 2015 Program: 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Wyatt SE, Basu P, Kruse CPS, Luesse DR. "BRIC20: Proteomics Analysis of Arabidopsis Seedlings in Microgravity." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 2015 Program: 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Basu P, Luesse DR, Wyatt SE. "BRIC20: Identification of Differentially Expressed Proteins in Seedlings Grown on the International Space Station." Presented at 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Program. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Wyatt SE. "Plant Gravitropism: novel approaches to understanding signal transduction." Keynote Lecture. Presented at Plant Biology Europe, EPSO/FESPB 2016 Congress, Prague, Czech Republic, June 26-30, 2016. Program. Plant Biology Europe, EPSO/FESPB 2016 Congress, Prague, Czech Republic, June 26-30, 2016. , Jun-2016
Articles in Other Journals or Periodicals	Luesse DR. "Plants in Space: Gravity Experiments with NASA." GARNish. 2016 Jun;25:10-3. , Jun-2016

Articles in Peer-reviewed Journals	Cook CA, Tucker A, Shen K, Wyatt SE. "Microarray identifies transcription factors potentially involved in gravitropic signal transduction in Arabidopsis." Gravit Space Res. 2015 Dec;3(2):20-9. http://gravitationalandspacebiology.org/index.php/journal/issue/view/55/showToc , Dec-2015